

# PROJECT TASK LIST

## Sistem SAPA – Layanan Akademik IPB University

Tim: Kelompok 4 | Mata Kuliah: Analisis & Desain Sistem | Fokus: Backend Development

| No                                | Task / Deliverable                                                                                                                                                                                                                               | Penanggung Jawab | Status   |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------|
| ❑ FASE 1 – ANALISIS & PERENCANAAN |                                                                                                                                                                                                                                                  |                  |          |
| 1                                 | <b>Wawancara &amp; Requirements Gathering</b><br><i>Wawancara stakeholder, pengumpulan kebutuhan fungsional &amp; non-fungsional</i>                                                                                                             | Semua Anggota    | ✔Selesai |
| 2                                 | <b>Analisis &amp; Perancangan Diagram Sistem</b><br><i>Use Case, Class Diagram, Activity Diagram, Sequence Diagram</i>                                                                                                                           | Semua Anggota    | ✔Selesai |
| 3                                 | <b>Perancangan Arsitektur Sistem &amp; ERD</b><br><i>Desain arsitektur FastAPI + React + PostgreSQL, perancangan skema database (ERD)</i>                                                                                                        | Backend Team     | ✔Selesai |
| ❑ FASE 2 – DESAIN UI/UX           |                                                                                                                                                                                                                                                  |                  |          |
| 4                                 | <b>Desain Low-Fidelity Wireframe</b><br><i>Sketsa alur dan layout halaman utama (login, dashboard, tiket, chatbot)</i>                                                                                                                           | Frontend Team    | ✔Selesai |
| 5                                 | <b>Desain Medium-Fidelity Mockup</b><br><i>Pengembangan wireframe menjadi mockup dengan struktur halaman yang lebih detail</i>                                                                                                                   | Frontend Team    | ✔Selesai |
| 6                                 | <b>Desain High-Fidelity Prototype (Figma)</b><br><i>Prototype interaktif lengkap dengan visual akhir di Figma; termasuk halaman Admin, Staff, dan Mahasiswa</i>                                                                                  | Frontend Team    | ✔Selesai |
| ❑ FASE 3 – PENGEMBANGAN BACKEND   |                                                                                                                                                                                                                                                  |                  |          |
| 7                                 | <b>Inisiasi Proyek &amp; Setup Environment</b><br><i>Setup repo Git, struktur proyek FastAPI, konfigurasi .env (SECRET_KEY, GOOGLE_CLIENT_ID, GEMINI_API_KEY, FERNET_KEY), CORS</i>                                                              | Backend Team     | ✔Selesai |
| 8                                 | <b>Pembuatan Model SQLAlchemy &amp; Migrasi Database</b><br><i>Model: User (Mahasiswa, StaffAkademik, AdminSistem), TiketLayanan, TanggapanStaff, Notifikasi, KnowledgeBase, ChatbotSession, AuditLog; migrasi via Alembic</i>                   | Backend Team     | ✔Selesai |
| 9                                 | <b>Pengembangan Modul Keamanan (security.py)</b><br><i>JWT Auth (HS256, 60 min), Google OAuth2 token verification, RBAC (cek_role), enkripsi kolom Fernet (EncryptedString), RSA-2048 key pair generation &amp; digital signature, audit log</i> | Backend Team     | ✔Selesai |

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| 10 | <b>Pengembangan API Autentikasi &amp; Otorisasi (auth.py)</b><br><i>POST /auth/login (Google OAuth2 → JWT), POST /auth/logout; validasi domain @apps.ipb.ac.id; auto-register user baru dengan role detection; pencatatan audit log</i>                                                                                 | Backend Team | ✔Selesai |
| 11 | <b>Pengembangan API Tiket Layanan (tiket.py)</b><br><i>POST / (multipart/form-data, upload KTM &amp; UKT, progressive profiling), GET / (daftar tiket by role), GET /{id} (detail + ownership check), PUT /{id} (update status by Staff/Admin), POST /{id}/tanggapan (balasan Staff + digital signature)</i>            | Backend Team | ✔Selesai |
| 12 | <b>Refaktor &amp; Validasi Schema Tiket (schemas.py)</b><br><i>Sinkronisasi kategori (Persuratan, Informasi), tambah field semester, validasi dinamis data_request per jenis surat (5 jenis: SKMA, Izin Akademik, Perubahan KRS, Rekomendasi Beasiswa, Magang) via @model_validator</i>                                 | Backend Team | ✔Selesai |
| 13 | <b>Pengembangan API Manajemen File (update.py)</b><br><i>POST /files/upload (validasi tipe PDF/JPG/PNG, max 5 MB, UUID filename), GET /files/{file_id} (download); class FileStorageService (singleton OOP)</i>                                                                                                         | Backend Team | ✔Selesai |
| 14 | <b>Pengembangan API Sistem Notifikasi (notifikasi.py)</b><br><i>GET /notifikasi/ (daftar per role), GET /notifikasi/unread-count, PATCH /notifikasi/read-all, PATCH /notifikasi/{id}/read; notifikasi otomatis tiket &amp; knowledge base; class NotifikasiService (OOP + static/class method)</i>                      | Backend Team | ✔Selesai |
| 15 | <b>Setup Vector Database FAISS &amp; Embedding (ingest.py)</b><br><i>Pipeline ingest PDF: PyPDFLoader → RecursiveCharacterTextSplitter (chunk 1000, overlap 200) → GoogleGenerativeAIEmbeddings (gemini-embedding-001) → FAISS; incremental update dengan checkpoint file; batch processing dengan rate-limit delay</i> | Backend Team | ✔Selesai |
| 16 | <b>Pengembangan API Chatbot RAG (chatbot.py)</b><br><i>POST /chatbot/tanya (RAG chain: FAISS retriever k=6 → Gemini 2.5 Flash, suhu 0.3, prompt SAPA Bot); GET /chatbot/riwayat; penyimpanan sesi ChatbotSession ke PostgreSQL; class SAPABotEngine (OOP)</i>                                                           | Backend Team | ✔Selesai |
| 17 | <b>Pengembangan API Knowledge Base Staff (knowledge_base.py)</b><br><i>GET /staff/knowledge-base/ (list + search), POST / (upload PDF pending, simpan ke ./data_pending), DELETE /{id} (hapus dokumen pending); validasi role Staff</i>                                                                                 | Backend Team | ✔Selesai |
| 18 | <b>Pengembangan API Sinkronisasi Knowledge Base Admin (admin_sync.py)</b>                                                                                                                                                                                                                                               | Backend Team | ✔Selesai |

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|    | GET /admin/sync/ (dashboard: pending/approved/rejected), POST /approve/{id} (pindah file → ./data, trigger FAISS ingest via BackgroundTasks, update status + timestamp + email admin), POST /reject/{id} (hapus file fisik, rekam jejak penolakan)                                                                                     |              |          |
| 19 | <b>Pengembangan API Keamanan &amp; Audit Log Admin (admin_security.py)</b><br>GET /admin/security/stats (statistik autentikasi & otorisasi), GET /admin/security/logs (audit log dengan pagination), POST /admin/tambah-user (registrasi user manual dengan validasi NIP)                                                              | Backend Team | ✓Selesai |
| 20 | <b>Pengembangan API Kelola Pengguna Admin (kelola_pengguna.py)</b><br>GET / (list semua user), POST / (tambah user baru, validasi role-spesifik: NIP/unit_kerja wajib untuk Staff), PUT /{email} (edit data + validasi instanceof), DELETE /{email} (soft delete: is_active=False), POST /{email}/reset-kunci (reset public key Staff) | Backend Team | ✓Selesai |
| 21 | <b>Pengembangan API Portal Staff – Kunci Keamanan (staff.py)</b><br>POST /staff/generate-key: generate RSA-2048 key pair, simpan public key ke DB, kirim private key (.pem) sebagai file download; cegah duplikasi kunci                                                                                                               | Backend Team | ✓Selesai |

#### □ FASE 4 – PENGEMBANGAN FRONTEND

|    |                                                                                                                                                                  |               |          |
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| 22 | <b>Setup Proyek React &amp; Routing</b><br>Inisiasi proyek React (Vite), konfigurasi routing, setup state management, integrasi Tailwind CSS                     | Frontend Team | ✓Selesai |
| 23 | <b>Halaman Autentikasi (Login Google OAuth)</b><br>Halaman login dengan tombol Google Sign-In, validasi domain @apps.ipb.ac.id, penyimpanan JWT token            | Frontend Team | ✓Selesai |
| 24 | <b>Dashboard &amp; Halaman Tiket Mahasiswa</b><br>Halaman buat tiket (form multipart dinamis per jenis surat, upload KTM/UKT), riwayat tiket, detail tiket       | Frontend Team | ✓Selesai |
| 25 | <b>Dashboard &amp; Antarmuka Staff Akademik</b><br>Daftar tiket masuk, form tanggapan (upload dokumen + private key .pem), manajemen knowledge base (upload PDF) | Frontend Team | ✓Selesai |
| 26 | <b>Dashboard &amp; Antarmuka Admin Sistem</b><br>Kelola pengguna (CRUD), sinkronisasi knowledge base (approve/reject), audit log & statistik keamanan            | Frontend Team | ✓Selesai |
| 27 | <b>Fitur Chatbot SAPA</b><br>Antarmuka chat interaktif, tampilan riwayat percakapan, integrasi dengan API /chatbot/tanya                                         | Frontend Team | ✓Selesai |

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| 28 | <b>Sistem Notifikasi Real-time (Frontend)</b><br><i>Badge unread count, panel notifikasi, mark as read (satu/semua), polling atau live update notifikasi</i> | Frontend Team | ✔Selesai |
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| ❑ FASE 5 – INTEGRASI & PENGUJIAN |                                                                                                                                                                                                                               |               |           |
| 29                               | <b>Integrasi API: Frontend ↔ Backend</b><br><i>Koneksi seluruh endpoint API ke antarmuka React; konfigurasi CORS; penanganan error &amp; loading state</i>                                                                    | Semua Anggota | ❑ Ongoing |
| 30                               | <b>Pengujian Unit &amp; Integrasi (QA)</b><br><i>Unit testing endpoint kritis (tiket, auth, chatbot); integration testing alur end-to-end (submit → notifikasi → tanggapan → digital signature); validasi Pydantic schema</i> | Semua Anggota | ❑ Ongoing |
| 31                               | <b>Perbaikan Bug &amp; Optimasi</b><br><i>Resolusi isu integrasi frontend-backend, perbaikan validasi form, optimasi query DB &amp; performa FAISS</i>                                                                        | Semua Anggota | ❑ Ongoing |

|                                     |                                                                                                                                                                                                              |               |                 |
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| ❑ FASE 6 – DEPLOYMENT & DOKUMENTASI |                                                                                                                                                                                                              |               |                 |
| 32                                  | <b>Deployment Sistem ke Cloud Server</b><br><i>Konfigurasi server (VPS/cloud), setup PostgreSQL production, konfigurasi environment variables, deployment FastAPI + React</i>                                | Backend Team  | ❑ Belum Dimulai |
| 33                                  | <b>Penyusunan Dokumentasi Sistem &amp; API</b><br><i>Dokumentasi arsitektur sistem, panduan instalasi &amp; konfigurasi, dokumentasi API (FastAPI /docs auto-generated + penjelasan manual per endpoint)</i> | Semua Anggota | ❑ Belum Dimulai |

Keterangan Status:

✔Selesai❑ Ongoing❑ Belum Dimulai✕Dibatalkan

**Catatan:** Task list ini disusun berdasarkan analisis menyeluruh terhadap kode sumber proyek (backend/). Nomor 7–21 merupakan penjabaran lengkap seluruh modul backend yang telah diimplementasikan, meliputi 10 router API, modul keamanan (JWT/OAuth/RSA/Fernet), pipeline RAG Chatbot, dan sistem sinkronisasi Knowledge Base berbasis AI.